



Lab 05: Loops

CSE 4108

Structured Programming I Lab



Lab Tasks

1. Reverse Digits 2 - Use of Loops:

In **Lab 4 - Expressions (Reverse Digits)**, you were asked to write a program that displays a *two-digit* number with its digits reversed. Generalize the program so that the input number can have any number of digits.

2. Even Universe:

Write a C program that continuously takes number as input from the user and stops only when the user enters an even number. Then the program displays the smallest odd number (last number does not count) so far. If the first input is even, the program displays NONE.

Sample run:

```
Enter a number: 31
Enter a number: 3
Enter a number: 45
Enter a number: 1
Enter a number: 5
Enter a number: 55
Enter a number: 4
Smallest odd number so far: 1
```

3. A lot to print:

Write a program that finds a bunch of things (***number of inputs, sum, average, maximum, number of odd values***) in a series of numbers entered by the user. The program must prompt the user to enter numbers one by one. When the user enters **0** or **a negative number**, the program will end and display the results.

Sample Run:

```
Enter a number: 5
Enter a number: 10
Enter a number: 11
Enter a number: 1
Enter a number: 3
Enter a number: 8
Enter a number: 10
Enter a number: 4
Enter a number: 1
Enter a number: 6
Enter a number: 7
Enter a number: 10
Enter a number: 7
Enter a number: 0
```

```
The number of inputs is 13
The sum of the input values is 83
The average of the input values is 6.38
The number of odd values input is 7
The maximum value entered was 11
```

4. Forever A loan:

ABC Bank Ltd. provides loans to its customers. The customers return the amount by a fixed monthly payment. On each month however, the remaining balance is incremented by the interest rate assigned for each customer. The total balance at the end of each month can be calculated using the following formula:

$$\text{currBalance} = \text{prevBalance} - \text{pay} + (\text{prevBalance} \times \text{rate})$$

Here,

currBalance = Updated Balance

prevBalance = Previous Balance

pay = Monthly Payment

rate = Monthly Interest Rate

Your task is to write a program that calculates the remaining payable amount on a loan after each monthly payment.

Your program will take the loan amount, yearly interest rate, and ask the user for monthly payments as inputs and show the remaining balance as outputs. When the user enters **0** or **a negative number**, the program will end.

You need to display each balance with two digits after the decimal point.

Sample Run:

Enter amount of loan: 20000.00

Enter interest rate: 6.0

Enter monthly payment: 386.66

Balance remaining after payment 1: \$19713.34

Enter monthly payment: 386.66

Balance remaining after payment 2: \$19425.25

Enter monthly payment: 0

End of Program!!

5. Captain's Shield:

Captain Bangladesh has a good shield, but he always wants the size of his shield to match his weight. After Eid-ul-Adha, the Captain has put on some weight and needs a new shield. The shield maker is fed up of the Captain's ever changing weight, so she wants a program that will output the design of the shield and make her life easier.

Write a program in C that would take an integer, representing the weight of the Captain as input and print out a shield like the sample output. The number representing the weight is always an integer.

Sample Run

Enter weight of Captain: 1

*

Enter weight of Captain: 2

*

Enter weight of Captain: 3

*

Hint: Try to understand the pattern of printing, count based on the input value, how many spaces and how many asterisks are there in each line. Once you derive the relation, the printing should be easy. You need to use loop in loop (nested loop) to achieve the desired output.

6. Fibonacci series:

In mathematics, the Fibonacci numbers form a sequence, called Fibonacci sequence, such that each number is the sum of the two preceding ones, starting from 0 and 1.

The sequence looks as follows:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

Write a C program that takes a positive integer, n, input and then prints the Fibonacci sequence upto the n^{th} value.

Sample run:

Enter sequence length: 5

Sequence: 0, 1, 1, 2, 3